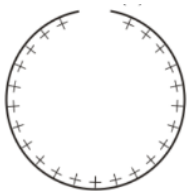
Lakshya JEE 2.0 (2025)

Physics

DPP: 5

Electric Charges and Fields

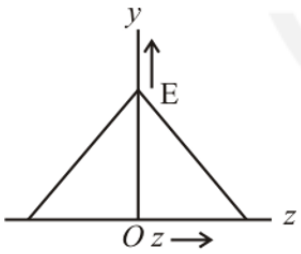
Q1 A ring of charge with radius $0.5\ \mathrm{m}$ has $0.002\pi\mathrm{m}$ gap. If the ring carries a charge of $+1\mathrm{C}$, the electric field at the center is



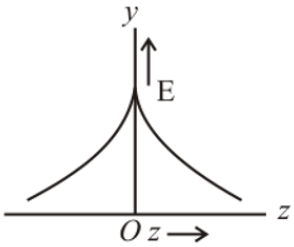
(A) $7.5 \times 10^7 \mathrm{NC}^{-1}$
(B) $7.2 \times 10^7 \mathrm{NC}^{-1}$
(C) $6.2 \times 10^7 \mathrm{NC}^{-1}$
(D) $6.5 \times 10^7 \mathrm{NC}^{-1}$

Q2 A circular ring carries a uniformly distributed positive charge and lies in the ***xy*** plane with center at the origin of the coordinate system. If at a point (0, 0, ***z***), the electric field is ***E***, then which of the following graphs is correct ?

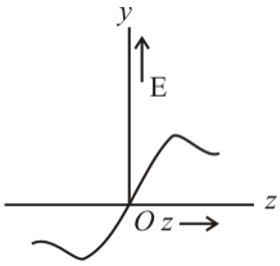
(A)



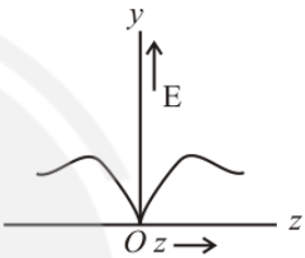
(B)



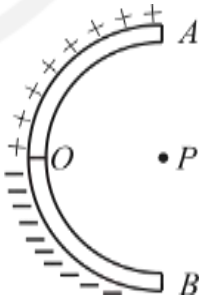
(C)



(D)

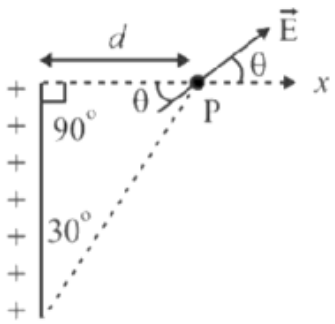


Q3 A thin glass rod is bent into a semicircle of radius r. A charge +Q is uniformly distributed along the upper half, and a charge −Q is uniformly distributed along the lower half, as shown in figure. The electric field E at P, the center of the semicircle, is



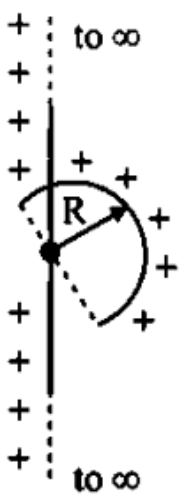
(A) $\frac{Q}{\pi^2 \varepsilon_0 r^2}$
(B) $\frac{2Q}{\pi^2 \varepsilon_0 r^2}$
(C) $\frac{4Q}{\pi^2 \varepsilon_0 r^2}$
(D) $\frac{Q}{4\pi^2 \varepsilon_0 r^2}$

Q4 The direction (θ) of $\vec{E}$ at point P due to uniformly charged finite rod will be



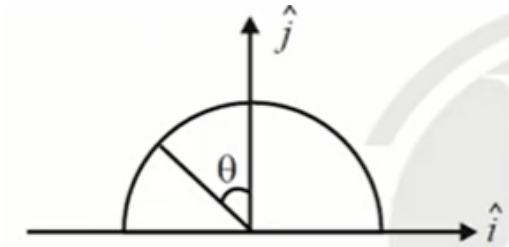
(A) 30° from the x-axis
(B) 45° from the x-axis
(C) 60° from the x-axis
(D) None of these

Q5 Find the force experienced by a semicircular wire having a charge q as shown in figure. Radius of the wire is and the line of charge with linear charge density λ passes through its center and is perpendicular to the plane of wire.



(A) $\frac{\lambda q}{2\pi^2\varepsilon_0 R}$
(B) $\frac{\lambda q}{\pi^2\varepsilon_0 R}$
(C) $\frac{\lambda q}{4\pi^2\varepsilon_0 R}$
(D) $\frac{\lambda q}{4\pi\varepsilon_0 R}$

Q6 A thin semi-circular ring of radius r has a positive charge q distributed uniformly over it. The net field at the centre O is:



(A) $\frac{q}{4\pi^2\varepsilon_0 r^2}\hat{j}$
(B) $-\frac{q}{4\pi^2\varepsilon_0 r^2}\hat{j}$
(C) $-\frac{q}{2\pi^2\varepsilon_0 r^2}\hat{j}$
(D) $\frac{q}{2\pi^2\varepsilon_0 r^2}\hat{j}$

Answer Key

Q1	(B)	Q4	(A)
Q2	(C)	Q5	(B)
Q3	(A)	Q6	(C)

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